

Course Title: Ethical Hacking and Vulnerability Assessment

Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)

	Unit-3: Vulnerability Assessment and Exploitation 3.1 Automated Vulnerability Scanning (OpenVAS, Nikto) 3.2 Manual Exploitation with Metasploit Framework 3.3 Web Application Security Testing (OWASP ZAP, Burp Suite) 3.4 Wireless Network Exploitation (Aircrack-ng, Kismet) 3.5 Privilege Escalation Techniques Unit-4: Post-Exploitation and Reporting 4.1 Maintaining Access and Covering Tracks 4.2 Password Cracking (John the Ripper, Hashcat) 4.3 Creating Custom Exploits with Python 4.4 Generating Penetration Testing Reports 4.5 Security Hardening and Patch Management Recommended Practical Exercises: • Setting Up a Pentesting Lab with Kali Linux • Performing Vulnerability Scanning with OpenVAS • Exploiting a Web Application Using OWASP ZAP • Wireless Network Cracking Using Aircrack-ng • Writing a Detailed Security Assessment Report
Reference Books	1. The Unofficial Guide to Ethical Hacking, Ankit Fadia, Macmillan Publishers India Ltd., 0333936795, 1st 2. Ethical Hacking 5-in-1 Book with Practicals,, Code Academy, 9788195765652, 2021 3. The Complete Ethical Hacking Book: A Comprehensive Beginner's Guide, , Orangebooks Publication, 935621204X, 1st 4. Cyber Safe Girl, Dr. Ananth Prabhu G, , 9789353820305, 5. An Ethical Guide to Hacking Mobile Phones, , Macmillan Publishers India Ltd., 1403928509, 6. Ethical Hacking: A Hands-on Introduction to Breaking In, Daniel G. Graham, No Starch Press, 9781718501874, 1st 7. Ethical Hacking: Techniques, Tools, and Countermeasures, , Jones & Bartlett Learning, 9781284248999, 4th 8. Hands-On Ethical Hacking and Network Defense, Michael T. Simpson, Nicholas Antill, Cengage Learning, 9780357509753, 4th 9. Certified Ethical Hacker (CEH) Version 12, EC-Council, EC-Council Academia, 9798885931052, 12th 10. Learn Ethical Hacking from Scratch, Zaid Sabih, Packt Publishing, 9781788622059, 1st 11. Incident Response & Computer Forensics by Jason Luttgens, Matthew Pepe, Kevin Mandia 12. "The Art of Memory Forensics" by Michael Hale Ligh et al. 13. "Malware Analyst's Cookbook and DVD" by Michael Hale Ligh et al 14. "Practical Malware Analysis" by Michael Sikorski and Andrew Honig 15. Official Documentation: Autopsy, Volatility, Wireshark, Cuckoo Sandbox, Ghidra
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.